

Neural Control of Robot Manipulators With Trajectory Tracking Constraints and Input Saturation

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Abstract—This article presents a control scheme for the robot manipulator's trajectory tracking task considering output error constraints and control input saturation. We provide an alternative way to remove the feasibility condition that most BLF-based controllers should meet and design a control scheme on the premise that constraint violation possibly happens due to the control input saturation. A bounded barrier Lyapunov function is proposed and adopted to handle the output error constraints. Besides, to suppress the input saturation effect, an auxiliary system is designed and emerged into the control scheme. Moreover, a simplified RBFNN structure is adopted to approximate the lumped uncertainties. Simulation and experimental results demonstrate the effectiveness of the proposed control scheme.

Index Terms—Barrier Lyapunov function (BLF), constrained control, input saturation, robot manipulator.

I. INTRODUCTION

CONSTRAINTS on output error commonly exist in engineering applications and are often interpreted as the performance requirement and safety specification [1], [2]. Constraint violation may degrade the control performance or even cause unexpected system damage. To tackle this problem, some constrained control methods have been investigated extensively these years, to name a few, the command governor [3], unbounded coordinate transformation method [4], and the nonovershooting control [5]. These methods either

modify the reference/control signals or transfer the constrained variables into new ones to modulate the final control input indirectly. Recently, the barrier Lyapunov function (BLF) has attracted more and more attention and acts as an effective tool in the control design for constrained nonlinear uncertain systems. The control law is derived directly from the Lyapunov stability analysis. Successful applications can be found in the prescribed tracking performance control of serial robot manipulators [6], handling the tracking error constraints effect on the marine surface vessel [7] and the full state-constrained control for nonlinear systems [8].

So far, many types of BLF have been proposed to handle the constraint effect, such as the log-type BLF [9], the integral-type BLF [8], and the tan-type BLF [10]. However, at the beginning of the control, a BLF-based controller usually needs prescribed time-varying transient performance constraints to achieve valid and stable control, as in [1] and [6]. Besides, to satisfy the constraints when the constrained variables limitlessly approach the predefined boundaries, the control input of a BLF-based controller tends to be infinite, which results in a paradox for the existence of the input saturation. Therefore, to handle the constraints, the BLF-based controller should meet the feasibility condition in [11], i.e., the constrained variable should always stay within the predefined boundaries. Parameters should also be appropriately designed, as it may be impractical if the constraints region is too small. Much research work has been made in [11] and [12] to remove the strict feasibility condition. In these works, the time derivative of the universal barrier function in [13] is regarded as an auxiliary system, which is subsequently incorporated into the control system. Then, the controller for the augmented system is designed under the dynamic surface control scheme. However, in authors' opinion, this method only partially relaxes the feasibility condition because of two points: 1) the initial system states still need to satisfy the constraints and 2) the control law will be discontinuous when the constrained variable equals to the constraints, and the case involving constraint violation has not been studied in the simulation. To further narrow the reality gap and to provide an alternative way to remove the feasibility condition radically, a new type of BLF should be designed and have the following properties: 1) be positive outside the constrained space and 2) the partial derivative with relative to the constrained variables be bounded at the whole space. To the best of our knowledge, there is

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no existing BLF being qualified for these conditions, which motivates part of the research in this article.

For constrained control, the input saturation effect, representing the capacity limitation of a plant, is a nontrivial source of the degradation of system performance or even system instability if not being properly handled with [14]. In [15], the input saturation effect is formulated as a model predictive control problem, and then, the proposed algorithms are adopted for estimating the upper bound of the control horizon, while this method is only suitable for linear time-invariant system. Model predictive control is widely used in the chemical industry and is able to handle input saturation. The control barrier function is also one of the methods to verify and enforce the safety properties of a plant. It is applied to the safety-critical control and unified with the Lyapunov function through quadratic programming to track the reference signals [16]. However, these methods are of high computational burden when the controlled plant is complex, nonlinear, and with uncertainties. To handle the output constraints and the input saturation simultaneously, an adaptive control framework would be a better choice. Wu and Xie [17] employed time-varying asymmetric BLFs, an auxiliary subsystem, and the Nussbaum gain technique to design an adaptive controller for uncertain nonlinear systems with time-varying full state constraints, input saturation, and unknown control direction. Yan *et al.* [18] introduced a nonlinear smooth function to tackle the saturation input nonlinearity, with HODOs compensating the unknown time-varying disturbances. In [19], the adaptive fuzzy output-feedback control is developed for a class of output-constrained uncertain nonlinear systems with input saturation and unmeasured states using a log-type BLF and an auxiliary system incorporating into the virtual control variable. This framework is refined and systematically illustrated in [14], where the input and output constraints are circumvented simultaneously. In the process of controller design, in order to obtain a smooth control law, the saturation function $\ell(\tau)$ is replaced with a hyper-tangent function $\tanh(\tau)$ in [14], and the auxiliary system is driven by the discrepancy between τ and $\tanh(\tau)$. However, the nonlinearity of $\tanh(\cdot)$, to some degree, may degrade the control performance since the states of the auxiliary system representing the influence of the input saturation will not converge to zero when the control input is not equal to zero.

This article considers both the output error constraints and the input saturation of robot manipulators and presents a trajectory tracking controller to fulfill these requirements simultaneously. Comparing with the existing results, the contributions of this article are summarized as follows.

- 1) A bounded BLF (BBLF) is proposed. Different from most of the existing BLF-based controllers, the proposed BBLF-based controller in this article is designed on the premise of taking the case when constraints violation occurs into consideration.
- 2) Inspired by [14], a modified auxiliary system is introduced to handle the input saturation effect, where the system states will converge to zeros if no control input saturation occurs.
- 3) A simplified RBFNN structure is adopted to greatly reduce the complexity of the network. Furthermore,

a simple criterion is proposed to quantify the contribution of the four items that constitute the proposed control law, so as to examine their respective effectiveness.

Notations are defined throughout this article: $\lambda_{\min/\max}(A)$ denotes the maximum/minimum eigenvalue of A ; $a \odot b = [a_1b_1, a_2b_2, \dots, a_nb_n]^T$ denotes the elementwise product between a and b ; and $|x|$ and $\|x\|$ denote the absolute and norm value of x , respectively. $\text{vec}(x_i), i = 1, 2, \dots, n$, denotes a column vector whose entries are x_i in sequence; $\text{diag}(x_i), i = 1, 2, \dots, n$, denotes a diagonal matrix whose diagonal elements are x_i with the rest entries being 0; and $\mathbf{0}$ denotes a compatible zero vector or zero square matrix.

II. PROBLEM FORMULATION AND PRELIMINARIES

A. Problem Formulation

Consider an n -link serial robot manipulator, its dynamics is governed by

$$H(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \ell(\tau) \quad (1)$$

where $H(q)$, $C(q, \dot{q})$, and $G(q)$ represent the inertia matrix, the Coriolis/centrifugal matrix, and the gravitation torque, respectively. q is the joint position with $\dot{q}$ being its corresponding velocity. In the following, $H(q)$, $C(q, \dot{q})$, and $G(q)$ are abbreviated for H , C , and G for saving space, and C and G are unknown.

1) *Input Saturation*: $\ell(\tau)$ embodies the input saturation effect with $\ell(\tau) = \varphi(\tau, \mu) + \Delta\tau$, where

$$\ell_i(\tau_i) = \begin{cases} \tau_i^{\max}, & \tau_i > \tau_i^{\max} \\ \tau_i, & \tau_i^{\min} \leq \tau_i \leq \tau_i^{\max} \\ \tau_i^{\min}, & \tau_i < \tau_i^{\min} \end{cases}$$

$$\varphi_i(\tau_i, \mu_i) = \begin{cases} \mu_i \tau_i^{\max} + \bar{\tau}_i^{\max}, & \tau_i > \mu_i \tau_i^{\max} \\ \tau_i, & \mu_i \tau_i^{\min} \leq \tau_i \leq \mu_i \tau_i^{\max} \\ \mu_i \tau_i^{\min} + \underline{\tau}_i^{\min}, & \tau_i < \mu_i \tau_i^{\min} \end{cases} \quad (2)$$

where τ_i , μ_i , $\ell_i(\cdot)$, and $\varphi_i(\cdot, \cdot)$ are the i th elements of τ , μ , $\ell(\cdot)$, and $\varphi(\cdot, \cdot)$, respectively, $\tau_i^{\max} > 0$ and $\tau_i^{\min} < 0$ are the upper and lower limits of the i th actuator, $\mu_i \in (0, 1)$ is a user-defined parameter, and

$$\bar{\tau}_i^{\max} = (1 - \mu_i) \tau_i^{\max} \tanh\left(\frac{\tau_i - \mu_i \tau_i^{\max}}{(1 - \mu_i) \tau_i^{\max}}\right)$$

$$\underline{\tau}_i^{\min} = (1 - \mu_i) \tau_i^{\min} \tanh\left(\frac{\tau_i - \mu_i \tau_i^{\min}}{(1 - \mu_i) \tau_i^{\min}}\right).$$

We notice that the piecewise function $\varphi(\tau, \mu)$, $\mu \in (0, 1)$, is twice differentiable relative to τ , and $\Delta\tau$ is bounded, which helps to get a smoother control input around the limits by designing a control law τ and then substituting it into $\varphi(\tau, \mu)$ rather than into $\ell(\tau)$. Note that, by using $\varphi(\tau, \mu)$ instead of $\bar{v} \times \tanh(\tau/\bar{v})$ employed in [14] with $\bar{v}$ being the limit of the actuator, the discrepancy $\Delta\tau$ between $\varphi(\tau, \mu)$ and $\ell(\tau)$ totally vanishes in $[\mu \odot \tau^{\min}, \mu \odot \tau^{\max}]$, and its maximum value can be tuned by μ .

2) *Output Error Constraints*: The output error constraints are defined as

$$\varpi := \{q \in \mathbb{R}^n \mid -k_{ci} < q_i - q_{di} < k_{ci}, i = 1, 2, \dots, n\} \quad (3)$$

where n is the degree of freedom of the robot manipulator, and k_{ci} are user-defined constraints. For simplicity, only symmetric constant constraints are considered in this article, while it is easy to deduce the time-varying and asymmetric constraints controller analogous to the proposed control framework.

The control problem to be resolved is stated as follows: design a trajectory tracking controller with a control law $\varphi(\tau, \mu)$ for an n -link robot manipulator in (1) under the output error constraints in (3) and the input saturation $\ell(\tau)$, such that the ultimately uniformly bounded (UUB) stability of the whole system is guaranteed and the tracking errors converge to a small bounded neighborhood around the origin.

B. Definition, Lemmas, and Assumptions

Assumption 1: The inertia matrix $H(q)$ of the robotic dynamics is exactly known, and the desired joint trajectory $q_d(t)$ is twice differentiable.

There are multiple techniques to guarantee system stability when analyzing the Lyapunov function, such as the logic-based switchings applied in [20] and the nonlinear measures in [21]. For the ease of control design, the following lemma is introduced.

Lemma 1 (See [22, Corollary 1.1 and Lemma 1.2]): Let the Lyapunov function candidate $V(\chi)$ be positive and continuously differentiable with a bounded initial value $V(\chi(0))$ and $\rho_c \|\chi\|^2 \leq V(\chi)$, where χ is a column vector consists of system states and weights of the neural networks, ρ_c is a positive scalar. If the following inequality holds:

$$\dot{V}(\chi) \leq -\gamma V(\chi) + \rho \quad (4)$$

where $\gamma > 0$ and $\rho > 0$, then $V(\chi)$ is bounded, and the solution of χ will converge exponentially to a compact set $\Omega = \{\chi \mid \|\chi\| \leq \sqrt{\rho/(\rho_c \gamma)}\}$.

Definition 1 (See [9, Definition 2]): A BLF is a scalar function $V_{\text{BLF}}(x)$ defined with respect to the system $\dot{x} = f(x)$ on an open region D containing the origin. BLF is continuous, positive definite, and has continuous first-order partial derivatives at every point of D . It has the property that $V(x) \rightarrow \infty$ as x approaches the boundary of D and satisfies $V(x(t)) \leq b \forall t \geq 0$ along the solution of $\dot{x} = f(x)$ for $x(0) \in D$ and some $b > 0$.

III. BOUNDED LYAPUNOV FUNCTION

Let us begin with an example. Consider a second order integrator system given as follows:

$$\begin{aligned} \dot{x}_1 &= x_2 \\ \dot{x}_2 &= f_2(x_1, x_2) + u \end{aligned} \quad (5)$$

where constraint is imposed on x_1 , requiring that $|x_1| < k_c$, and the Lyapunov function candidate is chosen as $V = V_{\text{BLF}}(x_1) + 0.5x_2^2$, where $V_{\text{BLF}}(x_1)$ is a BLF relative to x_1 . To guarantee system stability, the control law is designed as

$$u = -f_2(x_1, x_2) - k_2 x_2 - \tau_b \quad (6)$$

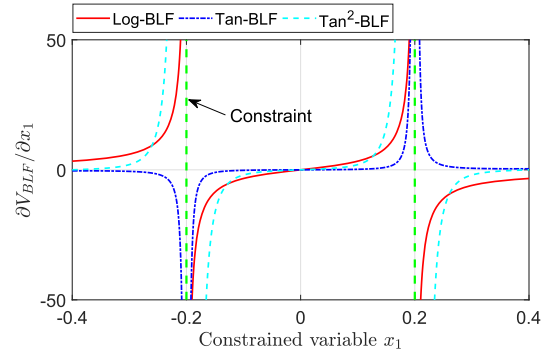


Fig. 1. Curves of $\partial V_{\text{BLF}}/\partial x_1$ derived from different types of BLFs.

where $\tau_b := \partial V_{\text{BLF}}/\partial x_1$ refers to the corresponding control term derived from a BLF.

Remark 1: Considering input saturation, the output error constraints are regarded as soft constraints, which implies that constraint violation should be taken into account during the control design. Fig. 1 shows the curves of τ_b derived from different types of BLFs in (7) from [9] and [10]

$$\begin{aligned} V_{\text{BLF}}^{\text{Log}} &= \frac{1}{2} \log\left(\frac{k_c^2}{k_c^2 - x_1^2}\right), \quad \tau_b^{\text{Log}} = \frac{x_1}{k_c^2 - x_1^2} \\ V_{\text{BLF}}^{\text{Tan}} &= \frac{k_c^2}{\pi} \tan\left(\frac{\pi x_1^2}{2k_c^2}\right), \quad \tau_b^{\text{Tan}} = \frac{x_1}{\cos^2\left(\frac{\pi x_1^2}{2k_c^2}\right)} \\ V_{\text{BLF}}^{\text{Tan}^2} &= \frac{k_c}{\pi} \tan^2\left(\frac{\pi x_1}{2k_c}\right), \quad \tau_b^{\text{Tan}^2} = \frac{\sin\left(\frac{\pi x_1^2}{2k_c^2}\right)}{\cos^3\left(\frac{\pi x_1^2}{2k_c^2}\right)}. \end{aligned} \quad (7)$$

Note that when $|x_1| > k_c$, $V_{\text{BLF}}^{\text{Log}}(x_1)$ is undefined, and $V_{\text{BLF}}^{\text{Tan}}(x_1)$ is less than 0. Although $V_{\text{BLF}}^{\text{Tan}^2}(x_1) > 0$ for $|x_1| > k_c$, $\tau_b^{\text{Tan}^2}$ behaves an opposite control direction, which may lead to system instability. Besides, all these τ_b s suffer from the singularity problem at $|x_1| = k_c$ if they are applied directly to handle the soft constraints. Therefore, BLFs in (7) cannot guarantee the system stability outside the constrained space. Furthermore, the infinite increase of the control torque around the constraints may also boost the saturation effect. Hence, we turn our attention to the magnitude and control direction of τ_b along $x_1 \in \mathbb{R}$ and circumventing the singularity problem of τ_b at $|x_1| = k_c$. Next, a new form of BLF will be designed whose curves of τ_b will have similar shape as other BLFs.

To handle the constraints' effect under the consideration of input saturation, we propose the BBLF whose definition is given as follows.

Definition 2: A BBLF is a scalar function $V_{\text{BBLF}}(x)$, $x \in \mathbb{R}$ defined with respect to the system $\dot{x} = f(x)$ along the real axis. BBLF is continuous, positive definite, and has first-order partial derivative. Let D be a close region and ∂D be the boundary of D . BBLF has the property that: 1) $\partial V_{\text{BBLF}}(x)/\partial x \rightarrow \tau_b^{\text{max}}$, as $x \rightarrow \partial D$ and 2) $\partial D = \arg \max_x |\tau_b(x)|$.

The following shows the specific mathematical form of BBLF:

$$\bar{V}(z_1) = \frac{1}{2} \sum_{i=0}^n V_{\text{BBLF}}(z_{1i}) + \frac{1}{2} z_1^T K_1 z_1$$

$$V_{\text{BBLF}}(z_{1i}) = k_i^{\text{des}} \frac{k_{\zeta i}^2 \zeta_i^2}{1 - k_{\zeta i}^2 \zeta_i^2}, \quad \zeta_i = \tanh\left(\kappa_i \frac{z_{1i}}{k_{ci}}\right) \quad (8)$$

where K_1 is a positive diagonal matrix, z_{1i} is an element of the constrained variable $z_1 \in \mathbb{R}^n$, k_{ci} is the user-defined constraint, $k_{\zeta i} \in (0, 1)$ is a user-defined scaling factor for ζ_i such that V_{BBLF} is bounded for $\forall z_{1i} \in \mathbb{R}$, and k_i^{des} is the control gain determined by (11). $(1/2)z_1^T K_1 z_1$ is an augmented component for the BBLF such that the magnitude of the feedback term will not vanished when $k_{ci} \rightarrow \infty$. Note that if $k_{\zeta i} = 1$ and κ_i is chosen wisely, then the proposed BBLF will behave as the traditional BLFs, i.e., as $z_{1i} \rightarrow k_{ci}$ and $\bar{V} \rightarrow \infty$. Next, taking the first time derivative of (8) yields

$$\dot{\bar{V}}(z_1) = (\tau_b^T + z_1^T K_1) \dot{z}_1 \quad (9)$$

where

$$\tau_b = K^{\text{des}} \text{vec}\left(h(\zeta_i) \frac{\zeta_i}{1 - k_{\zeta i}^2 \zeta_i^2}\right)$$

$$h(\zeta_i) = \frac{\kappa_i k_{\zeta i}^2}{k_{ci}} \frac{1 - \zeta_i^2}{1 - k_{\zeta i}^2 \zeta_i^2}$$

To meet the properties 1) and 2) of Definition 2, κ_i is selected according to $k_{\zeta i}$, such that τ_{bi} will reach its maximum when z_{1i} reaches constraint boundaries

$$\kappa_i = \ln\left(\sqrt{\frac{1 + \zeta_i}{1 - \zeta_i}}\right)$$

$$\zeta_i = \sqrt{\frac{3k_{\zeta i} - \sqrt{4 - 3k_{\zeta i}^2}}{2k_{\zeta i}}} \quad (10)$$

and $K^{\text{des}} = \text{diag}(k_i^{\text{des}})$

$$k_i^{\text{des}} = \tau_i^{\text{des}} \left(\frac{h(\zeta_i) \zeta_i}{1 - k_{\zeta i}^2 \zeta_i^2} \right)^{-1} \quad (11)$$

where τ_i^{des} is the user-defined parameter that decides the maximum control input generated from BBLF.

Remark 2: Fig. 2 presents the curves of τ_b^{BBLF} under different choices of k_{ζ} . For the convenience of expression, we define the region of z_1 , where $|\tau_b(z_1)| > 0.1\tau^{\text{des}}$ as the activate zone. We can see that τ_b mainly activates around the constraints at $z_1 = \pm k_c$. Besides, as $k_{\zeta} \rightarrow 1$, the active zone becomes narrower such that when z_1 converge to zero, and τ_b vanishes more quickly. The proposed BBLF can be applied on iterative learning control to guarantee the control accuracy of the final iteration, such as the motion control of a soft robot in [23], tracking control for strict-feedback systems in [24], and the policy iteration for human-robot coordination in [25], while the traditional BLFs are hard to realize that, at the first iteration of the iterative learning control, the control performance usually does not achieve the prescribed control performance. From Fig. 1, we can see that when the constrained variables are

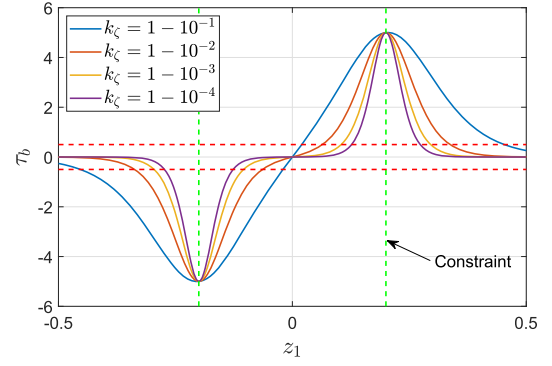


Fig. 2. Example of plots of $\tau_b^{\text{BBLF}} = \partial V_{\text{BBLF}}(z_1)/\partial z_1$ in (9) under different choices of k_{ζ} , where $\tau^{\text{des}} = 5$ N, and κ and k^{des} are selected according to (10) and (11). Red dashed lines denote $\tau_b = \pm 0.1\tau^{\text{des}}$, and green dashed lines denote $z_1 = \pm k_c$.

in violation of the constraints, system stability cannot be guaranteed because of the reverse control direction or the singularity problem at the constraint boundaries. Note that the curve of τ_b^{BBLF} is analogous to those in Fig. 1, except that τ_b^{BBLF} is bounded along the real axis and without singularity problem at the constraint boundary. Accordingly, the value of $V_{\text{BBLF}}(z_1)$ on the outside of constrained space is still positive. Therefore, the use of BBLF does not suffer from the feasibility condition. BBLF also features in offering flexibility of changing the boundedness of the τ_b via τ_b^{des} and the slope via k_{ζ} .

IV. CONTROLLER DESIGN

Unlike backstepping control in [26] where the controller is designed step by step, we introduce some auxiliary variables and give out the Lyapunov function candidate directly. Define the auxiliary variables q_r and r as follows:

$$q_r = \dot{q}_d + \Lambda e$$

$$r = \dot{e} + \Lambda e$$

where q_d denotes the desired joint trajectory; $e := q_d - q$ is the tracking error; and Λ represents a positive-definite diagonal matrix. Subsequently, we have $\dot{q} = q_r - r$ and $\ddot{q} = \dot{q}_r - \dot{r}$. In the field of adaptive control, there are many approximation tools to tackle unknown dynamics; in this article, the RBFNN that is time-efficient is employed, substituting which into (1) yields the following equality:

$$H\dot{r} = W^T S + \varepsilon + H\dot{q}_r - C z_2 - \ell(\tau) \quad (12)$$

where $W^T S + \varepsilon = C(q_r - \tilde{a}) + G$, $W^T S$ stands for the approximation of the lumped uncertainty, $\varepsilon \in L_\infty$ is the approximation error, and S is short for $S(q_r, q, \dot{q}, \tilde{a})$; $z_2 := r - \tilde{a}$, where $\tilde{a}$ is another auxiliary variable regarding the input saturation, and it will be specified later.

Let us take a further look at the approximation component $W^T S$; expanding the lumped uncertainty yields

$$C(q, \dot{q})q_r + G(q) - C(q, \dot{q})\tilde{a}$$

$$= \underbrace{C(q_d - e, \dot{q}_d - \dot{e})(\dot{q}_d + \Lambda e) + G(q_d - e)}_{\approx \hat{W}_1^T S_1} - \underbrace{C(q, \dot{q})\tilde{a}}_{\approx \hat{W}_2^T S_2} \quad (13)$$

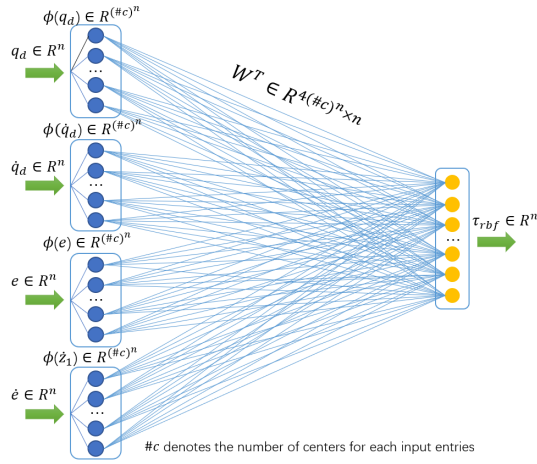


Fig. 3. Illustration of the simplified structure of the RBFNN $\hat{W}_1^T S_1(q_d, \dot{q}_d, e, \dot{e})$, where the yellow solid circles denote the output neurons and the blue solid circles stand for the hidden neurons mapping the input column vector into a scalar by using a Gaussian function in (14).

We divide the abovementioned lumped uncertainty into two parts and use two RBFNNs to approximate them separately: $\hat{W}_1^T S_1(q_d, \dot{q}_d, e, \dot{e}) \approx C(q_d - e, \dot{q}_d - \dot{e})(\dot{q}_d + \Lambda e) + G(q_d - e)$ and $\hat{W}_2^T S_2(q, \dot{q}, \tilde{a}) \approx -C(q, \dot{q})\tilde{a}$. Hence, $S = [S_1^T, S_2^T]^T$, and $\hat{W}^T = [\hat{W}_1^T, \hat{W}_2^T]$.

Remark 3: To reduce the number of the hidden neurons, a simplified structure of the RBFNN for $\hat{W}_1^T S_1$ is adopted, as shown in Fig. 3, where $\hat{W}_2^T S_2(q, \dot{q}, \tilde{a})$ shares a similar structure, $S_1 = [\Phi_{q_d}(q_d)^T, \Phi_{\dot{q}_d}(\dot{q}_d)^T, \Phi_e(e)^T, \Phi_{\dot{e}}(\dot{e})^T]^T$, $S_2 = [\Phi_q(q)^T, \Phi_{\dot{q}}(\dot{q})^T, \Phi_{\tilde{a}}(\tilde{a})^T]^T$, and the Gaussian function is

$$\phi(x) = \exp\left(-\frac{(x - c)^T (\Sigma^2)^{-1} (x - c)}{2}\right) \quad (14)$$

where c is the center, x is the input signal, and Σ is the standard deviation matrix. Although the simplified network may results in larger $\|\varepsilon\|$, it can greatly reduce the complexity of the network. Simulation results show that the degradation of the control performance is acceptable. Besides, we select the input signals of the first RBFNN as $q_d, \dot{q}_d, e, \dot{e}$ rather than the commonly used $q, \dot{q}, q_r$ out of the consideration that at the transient phase, magnitudes of q and $\dot{q}$ are unpredictable, the centers will need some gut feeling to be set, and these parameters may varies with different desired trajectories. However, with the known q_d and $\dot{q}_d$, the centers can be set to make part of the network sufficiently excited under limited neurons. The second RBFNN uses q (not q_d and e), $\dot{q}$ (not $\dot{q}_d$ and $\dot{e}$), and $\tilde{a}$ as input signals in order to inject more information into the network for better approximation.

Let $z_1 = e$ and define an error dynamic system governed by the auxiliary variables mentioned earlier and the robot dynamics in (1) as follows:

$$\begin{aligned} \dot{z}_1 &= z_2 + \tilde{a} - \Lambda z_1 \\ \dot{z}_2 &= \dot{r} - \ddot{a}. \end{aligned} \quad (15)$$

The Lyapunov function candidate is then chosen to be

$$V = \bar{V} + \frac{1}{2} z_2^T H z_2 + \frac{1}{2} \text{tr}(\tilde{W}^T \Gamma^{-1} \tilde{W}) \quad (16)$$

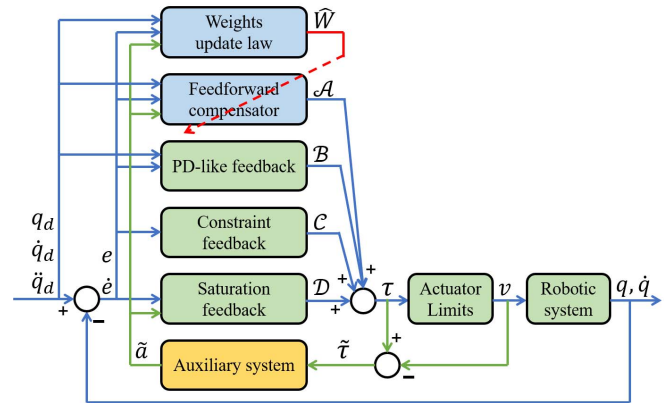


Fig. 4. Block diagram of the proposed control scheme.

where $\tilde{W} = W - \hat{W}$. Taking its first time derivative along (9), (12), and (15) and the skew-symmetric property of $\dot{H} - 2C$, one can obtain

$$\begin{aligned} \dot{V} &= (\tau_b^T + z_1^T K_1)(z_2 + \tilde{a} - \Lambda z_1) - \text{tr}(\tilde{W}^T \Gamma^{-1} \dot{\tilde{W}}) \\ &\quad + z_2^T (W^T S + \varepsilon + H \dot{q}_r - \ell(\tau) - H \dot{\tilde{a}}). \end{aligned} \quad (17)$$

Design the following auxiliary system:

$$\dot{\tilde{a}} = -H^{-1}(K_{\tilde{a}} \tilde{a} + \tilde{\tau}) \quad (18)$$

where $\tilde{\tau} = \tau - \varphi(\tau, \mu)$ so that the input saturation effect can be properly handled in the stability analysis. Finally, the control scheme is proposed as follows:

$$\begin{aligned} \tau &= \underbrace{\hat{W}^T S + H \dot{q}_r}_A + \underbrace{K_2 r + K_1 z_1}_B + \underbrace{\tau_b}_C + \underbrace{(K_{\tilde{a}} - K_2) \tilde{a}}_D \\ v &= \varphi(\tau, \mu) \end{aligned} \quad (19)$$

with the updated law

$$\dot{\hat{W}} = \Gamma (S z_2^T - \gamma \hat{W}) \quad (20)$$

where the proposed controller consists of four components, as shown in Fig. 4: 1) $\mathcal{A}$ serves for nonlinear compensation; 2) $\mathcal{B}$ is to suppress the compensation error and to improve the system's robustness; 3) $\mathcal{C}$ is responsible for handling the constraint problem; and 4) $\mathcal{D}$ systematically incorporates the input saturation effects into the controller; v is the final designed control input.

Remark 4: It worth mentioning the benefit of the boundedness of τ_b . Combined with τ in (19), the proposed auxiliary system in (18) is bounded-input bounded-output (BIBO) since: 1) the user-defined $K_{\tilde{a}}$ and the inverse of inertia matrix H are positive definite throughout the control, which implies that the auxiliary system is Hurwitz and 2) τ_b generated from BBLF is bounded regardless of the value of z_1 , and thus, the input for the auxiliary system is also bounded. Therefore, $\tilde{a} \in L_\infty$ during the control.

Theorem 1: For an n -link serial robot manipulator with dynamics in (1) under Assumption 1, unknown C and G , the control input saturation in (2), and the output error constraints in (3), if it is controlled by (19) with an auxiliary system given by (18) and the updated law in (20), then the

TABLE I
PARAMETERS OF THE RBFNNs FOR SIMULATIONS

inputs	c	Σ
$q_d \in \mathbb{R}^2$	$[0.05, 0, 0.25] \times [0.05, 0, 0.25]$	$diag(0.7, 0.9)$
$\dot{q}_d \in \mathbb{R}^2$	$[-0.1w, 0, 0.1w] \times [-0.1w, 0, 0.1w]$	$diag(1.3, 1.8)$
$e \in \mathbb{R}^2$	$[-0.08, 0, 0.08] \times [-0.06, 0, 0.06]$	$diag(0.7, 0.7)$
$\dot{e} \in \mathbb{R}^2$	$[-0.15, 0, 0.15] \times [-0.2, 0, 0.2]$	$diag(15.2, 15.4)$
$q \in \mathbb{R}^2$	$[0.05, 0, 0.25] \times [0.05, 0, 0.25]$	$diag(0.7, 0.9)$
$\dot{q} \in \mathbb{R}^2$	$[-0.1w, 0, 0.1w] \times [-0.1w, 0, 0.1w]$	$diag(1.3, 1.8)$
$\ddot{a} \in \mathbb{R}^2$	$[-1.5, 0, 1.5] \times [-1.5, 0, 1.5]$	$diag(5.0, 5.0)$

whole system will be UUB stable, and the tracking errors of the robot manipulator will converge to a small region around the origin.

Proof: The proof can be found in the Appendix. $\square$

V. SIMULATION STUDIES

A. Simulation Settings

A two-link robot manipulator in [27] is adopted to verify the proposed control scheme, whose dynamics equation is

$$\begin{aligned}
 H &= \begin{bmatrix} p_1 + 2p_2 \cos(q_2) & p_3 + p_2 \cos(q_2) \\ p_3 + p_2 \cos(q_2) & p_3 \end{bmatrix} \\
 C &= \begin{bmatrix} -p_2 \sin(q_2) \dot{q}_2 & -p_2 \sin(q_2) (\dot{q}_1 + \dot{q}_2) \\ p_2 \dot{q}_1 & 0 \end{bmatrix} \\
 G &= \begin{bmatrix} (m_1 + m_2)l_1 \cos(q_2) + m_2 l_2 \cos(q_1 + q_2) \\ m_2 l_2 \cos(q_1 + q_2) \end{bmatrix} g \quad (21)
 \end{aligned}$$

where $p_1 = (m_1 + m_2)l_1^2 + m_2 l_2^2$, $p_2 = m_2 l_1 l_2$, $p_3 = m_2 l_2^2$, $g = 9.8 \text{ m/s}^2$, $l_1 = 1.0 \text{ m}$, $l_2 = 0.8 \text{ m}$, and $m_1 = m_2 = 0.5 \text{ kg}$.

Parameters of the proposed controller are given: for the input saturation effect in (2), $\tau^{\max}/\tau^{\min} = \pm[30, 15]^T \text{ N} \cdot \text{m}$ and $\mu = [0.92, 0.92]^T$; for the constraints effect in (8) and (3), $k_c = [0.02, 0.02]^T$, $k_\zeta = [0.995, 0.995]^T$, $\tau_b^{\text{des}} = [20, 10]^T$, and $K_1 = \text{diag}(40, 30)$, wherein κ_i and k_i^{des} are obtained according to (10) and (11), respectively; $\Lambda = \text{diag}(9, 12)$ in (15); for the auxiliary system (18), $K_{\bar{a}} = \text{diag}(10, 12)$; and in (19), $K_2 = \text{diag}(30, 8)$. Parameters of the two RBFNNs and the updated law in (20) are selected as $S_1 = [\phi_{q_d}^T(q_d), \phi_{\dot{q}_d}^T(\dot{q}_d), \phi_e^T(e), \phi_{\dot{e}}^T(\dot{e})]^T \in \mathbb{R}^{36}$ and $S_2 = [\phi_q^T(q), \phi_{\dot{q}}^T(\dot{q}), \phi_{\ddot{a}}^T(\ddot{a})]^T \in \mathbb{R}^{27}$ whose centers and standard deviations are given in Table I; $\gamma = 0.001$, and $\Gamma = \text{diag}(1.2\mathbf{I}_2, 1.2\mathbf{I}_2, 1.0\mathbf{I}_2, 1.0\mathbf{I}_2, 0.2\mathbf{I}_2, 0.2\mathbf{I}_2, 0.8\mathbf{I}_2) \in \mathbb{R}^{14 \times 14}$ where $\mathbf{I}_2$ denotes a 2×2 identity matrix; and $\hat{W}_1^T(0) = \mathbf{0} \in \mathbb{R}^{2 \times 36}$ and $\hat{W}_2^T(0) = \mathbf{0} \in \mathbb{R}^{2 \times 27}$.

Remark 5: There are only three parameters of BBLF to be chosen for practical application. k_c is predefined by the task. k_ζ determines the slope of τ_b , and τ_b^{des} decides the boundedness of τ_b . The closer k_ζ approaches to 1, the smaller the magnitude of τ_b , when the constrained variable stays away from the constraints. The recommended choice of k_ζ ranges from 0.99 to 1×10^{-4} , τ_b^{des} being half of the limits of the actuator. Trial and error is still needed to find a suitable set of control parameters.

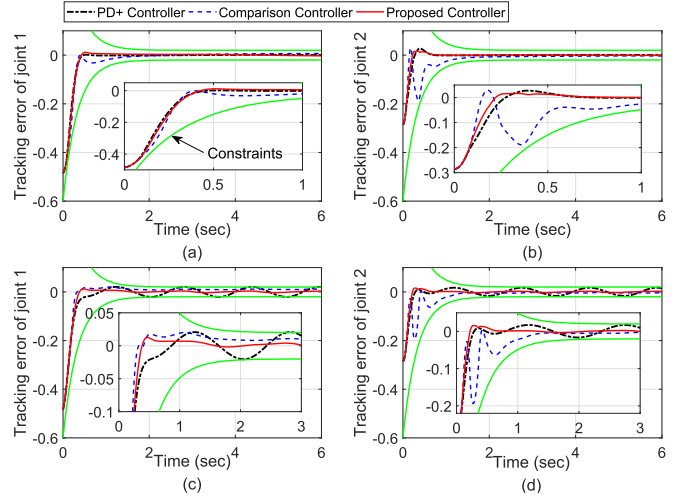


Fig. 5. Tracking performance under different reference trajectories ($\tau^{\max} = [30, 15]^T \text{ N}$). (a) and (b) $\omega = 1.0 \text{ rad/s}$. (c) and (d) $\omega = 4.0 \text{ rad/s}$.

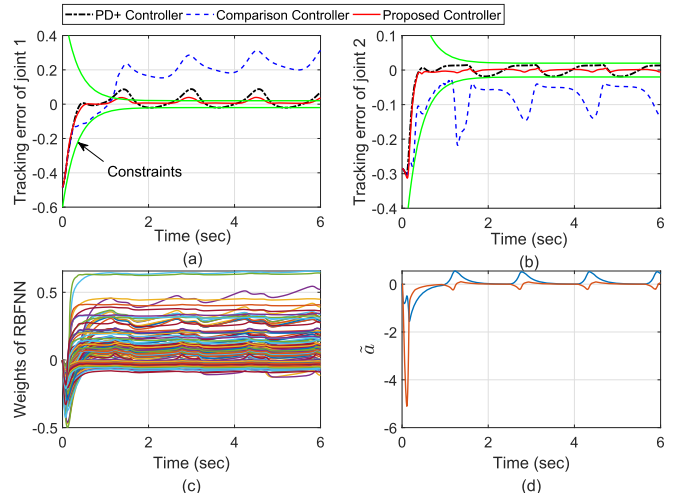


Fig. 6. Tracking performance under reference trajectories ($\omega = 4.0 \text{ rad/s}$ and $\tau^{\max} = [19, 8]^T \text{ N}$). (a) and (b) Tracking errors of joint 1 and joint 2. (c) and (d) Weights for the RBFNN and the auxiliary system's output signals in the proposed controller.

The control performance of the proposed control scheme is compared with the controller in [14] (refers to “Comparison controller”) and the PD+ controller given as follows:

$$v_{pd} = \ell(K_p e + K_d \dot{e} + G) \quad (22)$$

where $K_p = \text{diag}(320, 150)$, $K_d = \text{diag}(50, 15)$, $\tau^{\max}/\tau^{\min} = \pm[30, 15]^T$, and the gravitation torque is fully compensated by G .

The robot manipulator is commanded to track two reference trajectories with different frequencies using the same set of control parameters. The desired trajectory is

$$q_d = [0.2 \sin(\omega t) + 0.1, 0.2 \cos(\omega t) + 0.1]^T \quad (23)$$

where ω denotes the frequency of the desired trajectory. The effectiveness of the proposed controller will be examined by three cases.

- 1) *Case One:* The tracking performance will be examined under $\omega = 1.0$ and 4.0 rad/s separately, and

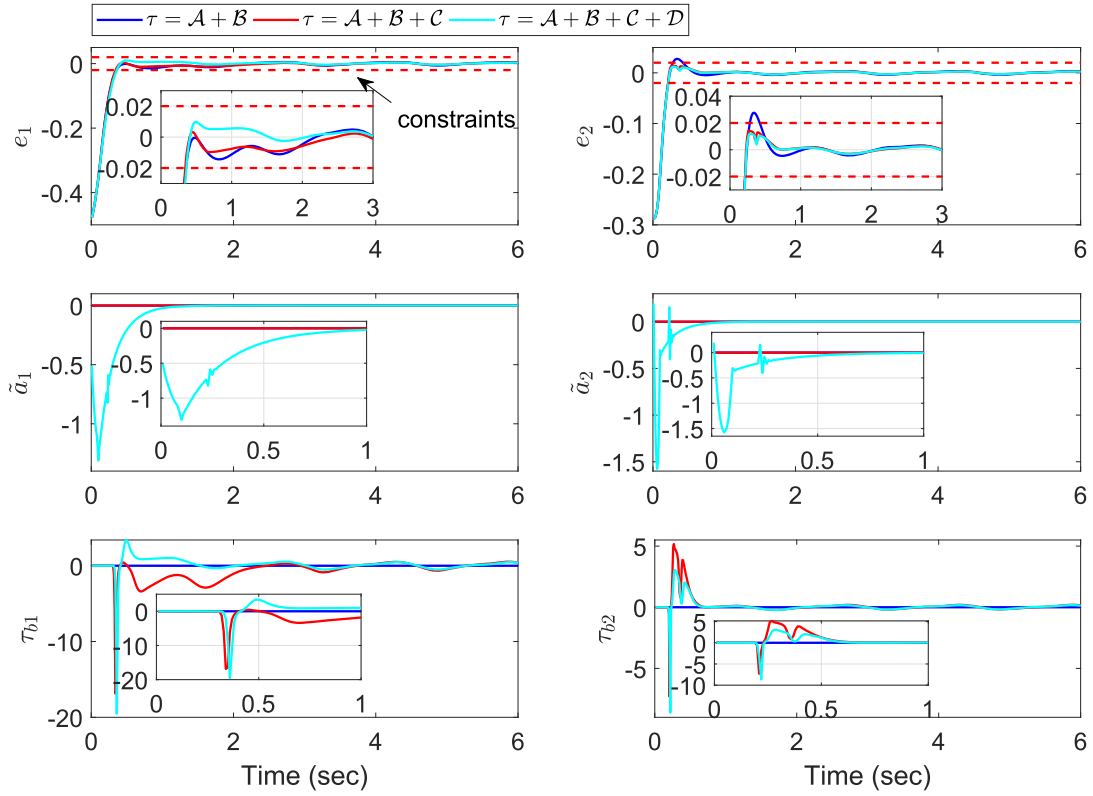


Fig. 7. **Case Two:** results of tracking the reference trajectory with $w = 4.0$ rad/s under three different combinations of the four components of the proposed controller. Left: tracking results for joint 1. Right: tracking results for joint 2.

the convergence of $\hat{W}_1$ and $\hat{W}_2$ will be checked. Note that the parameters of the proposed controller will remain unchanged under different choices of w , only except the choices of centers of $\dot{q}_d$ and $\dot{q}$ in Table I.

- 2) *Case Two:* We will take a further research at the control performance at $w = 4.0$ rad/s, where the contributions of the four components $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ in (19) are discussed and quantified by the following equations:

$$\begin{aligned} \mathcal{P}_i^{\mathcal{A}} &= |\mathcal{A}_i|/\mathcal{T}_i, & \mathcal{P}_i^{\mathcal{B}} &= |\mathcal{B}_i|/\mathcal{T}_i \\ \mathcal{P}_i^{\mathcal{C}} &= |\mathcal{C}_i|/\mathcal{T}_i, & \mathcal{P}_i^{\mathcal{D}} &= |\mathcal{D}_i|/\mathcal{T}_i \\ \mathcal{T}_i &= |\mathcal{A}_i| + |\mathcal{B}_i| + |\mathcal{C}_i| + |\mathcal{D}_i|. \end{aligned} \quad (24)$$

This evaluation index only shows the tendency of the control torque, as $\mathcal{D}$ does not exactly reflect the control torque relating to the input saturation.

- 3) *Case Three:* The tracking performance when constraint violation happens will be analyzed, by means of deliberately decreasing the absolute value of $\tau^{\max}/\tau^{\min}$, i.e., the saturation of the actuator.

B. Simulation Results

1) *Case One:* Fig. 5 shows the tracking performance of the controllers. It is obvious that the proposed controller guarantees the tracking errors within the constraints and

has much smaller tracking errors than the PD+ controller and the comparison controller, which demonstrates its superiority. Besides, the bounded errors are also in line with the results of stability analysis in the Appendix. Fig. 6(c) shows the convergence of the weights, which, to some extent, proves the approximation capability of the simplified RBFNN structure.

2) *Case Two:* The effectiveness of handling the constraints and input saturation effect is demonstrated in Fig. 7 by comparing the tracking errors when masking $\mathcal{C}$ and $\mathcal{D}$ (i.e., let $\mathcal{C} + \mathcal{D} = \mathbf{0}$) and $\mathcal{D}$ (i.e., let $\mathcal{D} = \mathbf{0}$) of τ in (19), respectively. From Fig. 7, we can notice that when $\mathcal{C}$ and $\mathcal{D}$ is masked, the output error constraint violation of a joint 2 happens. Furthermore, when we compare the tracking errors of case $\tau = \mathcal{A} + \mathcal{B} + \mathcal{C}$ and case $\tau = \mathcal{A} + \mathcal{B} + \mathcal{C} + \mathcal{D}$, we can clearly find that the addition of $\mathcal{D}$ improves the controller's tracking performance. These results motivate us to analyze the contribution of each component at the transient phase of the control, which is quantified by (24) and showed in Fig. 8. From this figure, we can see that at the beginning of the control ($t = 0 \sim 0.5$ s), $\mathcal{B}$ dominates the control input, while, as the tracking errors decrease, $\mathcal{P}^{\mathcal{B}}$ becomes smaller in contrast to an increase of $\mathcal{P}^{\mathcal{A}}$. During this transient period, $\mathcal{P}^{\mathcal{D}}$ climbs to its maximum to suppress the undesired saturation effect. Similarly, the contribution of $\mathcal{C}$ mainly focuses on the time interval when the tracking error approximates the predefined constraints. As the system becomes stable,

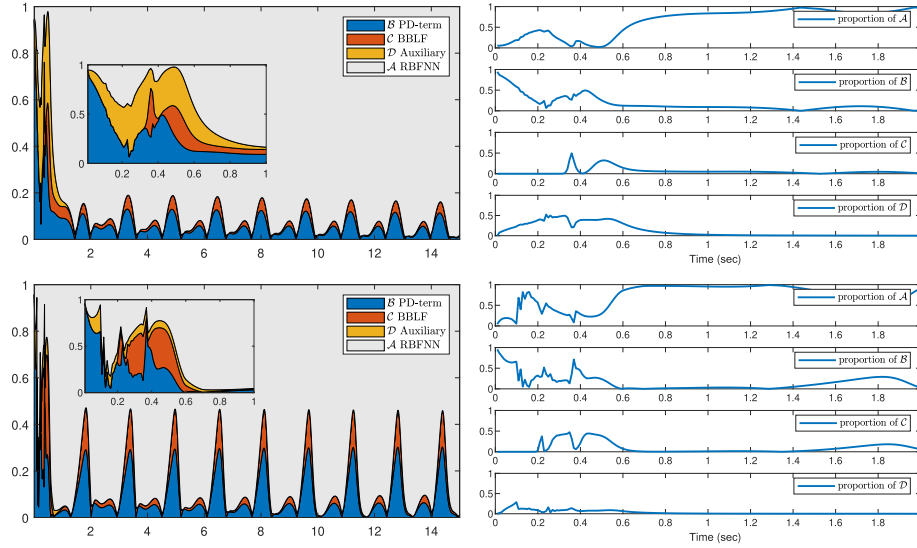


Fig. 8. **Case Two:** time series of the proportion for the contribution of each components in (19) when tracking with the reference trajectory of $w = 4.0$ rad/s under $\tau = \mathcal{A} + \mathcal{B} + \mathcal{C} + \mathcal{D}$. Left: stack graphs of the proportion for the contribution of $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$, where the upper subfigure shows the case of joint 1 and the lower one shows the case of joint 2. Right: charts of specific proportion for each component in its left subfigure at $t = 0 \sim 2$ s.

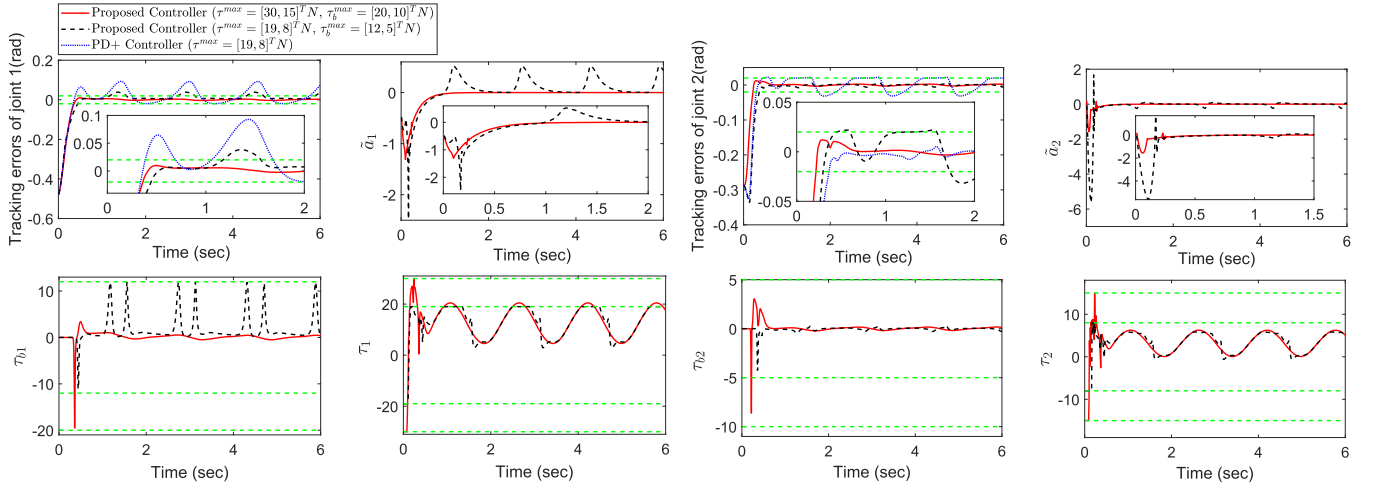


Fig. 9. **Case Three:** Detailed control performance of the proposed controller when constraint violation happens.

the output of the neural networks $\mathcal{A}$ takes the dominant place of the control input. Furthermore, as the saturation effect vanishes, $\mathcal{P}^D$ finally equals to zeros. These observations are in line with the theoretical control design, since each component has functioned well in tackling its corresponding effect.

3) *Case Three:* To further verify the distinct property of the proposed BBLF function, i.e., constraint violation does not affect the system's stability and the smoothness of the control input, we deliberately change the saturation of the actuators into $\tau^{\max}/\tau^{\min} = [19, 8]^T$ N·m such that constraints violation is bound to happen periodically, which can be seen in the tracking errors of the PD+ controller (see the dotted blue line in Fig. 9). From the tracking errors in Fig. 6, we can see that the comparison controller fails to stabilize the system when the constraint violation happens. In Fig. 9, although constraint violation incurs the saturation problem and the periodic activation of $\mathcal{C}$, with the proposed controller, the system is still stable and has smaller tracking errors than

those of the PD+ controller. These observations support the theoretical stability analysis of the proposed control scheme.

VI. EXPERIMENTAL STUDIES

To further verify the validity of the proposed control scheme, some experiments are carried out on the Baxter robotic platform. Its left arm is of 7-DOF named "left s_0 ," "left s_1 ," "left e_0 ," "left e_1 ," "left w_0 ," "left w_1 ," and "left w_2 " from the base to the end-effector. The left arm of the Baxter is commanded to track predefined trajectories with varying payloads, as shown in Fig. 10. The scenarios that the left robot manipulator performing trajectory tracking task grasping nothing, half bottle of water, and a full bottle of water are referred as Case 1, Case 2, and Case 3, respectively. The initial joint positions of the left arm in these cases are $-\pi/9$, $-\pi/4$, 0 , $\pi/2+0.1$, 0 , $\pi/4-0.1$, and 0 rad in sequence. For the sake of simplicity and without generality, "left e_1 " (joint 1) and "left w_1 " (joint 2) are selected to carry out the tasks, whose reference trajectories are given in (25), while



Fig. 10. Illustration of the experimental setup.

TABLE II
PARAMETERS OF THE RBFNNs FOR EXPERIMENTS

input	c	Σ
$q_d \in \mathbb{R}^2$	$[1.27, 1.37, 1.47] \times [0.68, 0.78, 0.98]$	$\text{diag}(0.4, 0.5)$
$\dot{q}_d \in \mathbb{R}^2$	$[-0.4, 0, 0.4] \times [-0.4, 0, 0.4]$	$\text{diag}(1.3, 1.8)$
$e \in \mathbb{R}^2$	$[-0.15, -0.05, 0, 0.05, 0.15] \times [-0.12, -0.04, 0, 0.04, 0.12]$	$\text{diag}(0.25, 0.25)$
$\dot{e} \in \mathbb{R}^2$	$[-0.25, -0.1, 0, 0.1, 0.25] \times [-0.2, -0.08, 0, 0.08, 0.2]$	$\text{diag}(0.5, 0.7)$
$q \in \mathbb{R}^2$	$[1.27, 1.37, 1.47] \times [0.68, 0.78, 0.98]$	$\text{diag}(0.4, 0.5)$
$\dot{q} \in \mathbb{R}^2$	$[-0.4, 0, 0.4] \times [-0.4, 0, 0.4]$	$\text{diag}(1.3, 1.8)$
$\tilde{a} \in \mathbb{R}^2$	$[-0.6, -0.2, 0, 0.2, 0.6] \times [-0.6, -0.2, 0, 0.2, 0.6]$	$\text{diag}(0.7, 0.7)$

positions of other joints remain unchanged. The initial error of the robot manipulator is $e(0) = [-0.1, 0.1]^T$ rad

$$q_d = [\pi/2 + 0.2 \cos(4t) - 0.2, \pi/4 + 0.2 \sin(4t)]^T. \quad (25)$$

Parameters of the controller in the experiments are as follows: $\mu = [0.92, 0.92]^T$, $k_c = [0.07, 0.07]^T$, $k_\zeta = [0.99, 0.995]^T$, $\tau_b^{\text{des}} = [3, 1]^T$, $K_1 = \text{diag}(10, 4)$, $\Lambda = \text{diag}(9, 5)$, $K_{\tilde{a}} = \text{diag}(10, 3)$, and $K_2 = \text{diag}(4, 1.2)$. Parameters of the two RBFNNs and the updated law in (20) are selected as follows: $S_1 \in \mathbb{R}^{68}$ and $S_2 \in \mathbb{R}^{43}$ whose centers and standard deviations are given in Table II; $\gamma = 0.001$, $\Gamma = \text{diag}(0.25\mathbf{I}_2, 0.25\mathbf{I}_2, 0.25\mathbf{I}_2, 0.25\mathbf{I}_2, 0.1\mathbf{I}_2, 0.1\mathbf{I}_2, 0.1\mathbf{I}_2) \in \mathbb{R}^{14 \times 14}$, $\hat{W}_1^T(0) = \mathbf{0} \in \mathbb{R}^{2 \times 68}$, and $\hat{W}_2^T(0) = \mathbf{0} \in \mathbb{R}^{2 \times 43}$. Parameters of the PD+ controller are chosen as $K_p = \text{diag}(60, 10)$ and $K_d = \text{diag}(4, 1.2)$. Experiments are performed using the two controllers with the control rate at 100 Hz and joint positions' sample rate at 100 Hz, and the weights of the RBFNN are updated at 50 Hz.

Figs. 11 and 12 show the tracking errors of the PD+ and the proposed controller with or without input saturation. It is

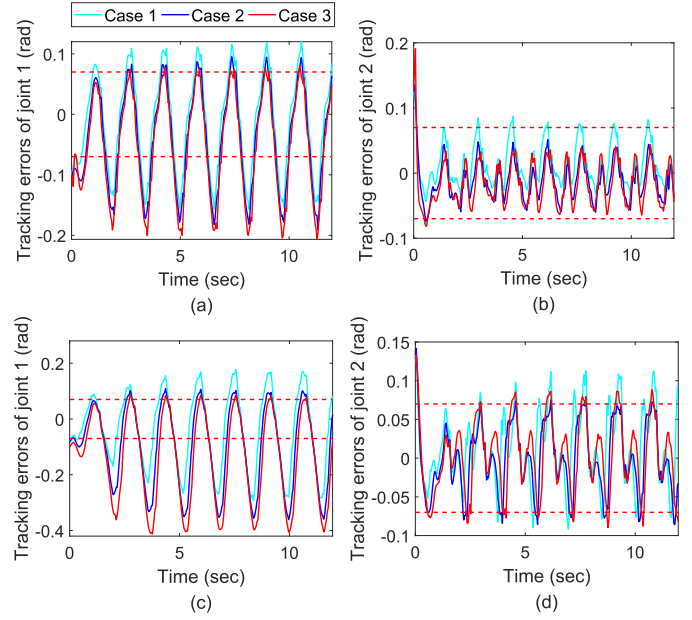


Fig. 11. Tracking performance of the PD+ controller. (a) and (b) Without input saturation. (c) and (d) With input saturation.

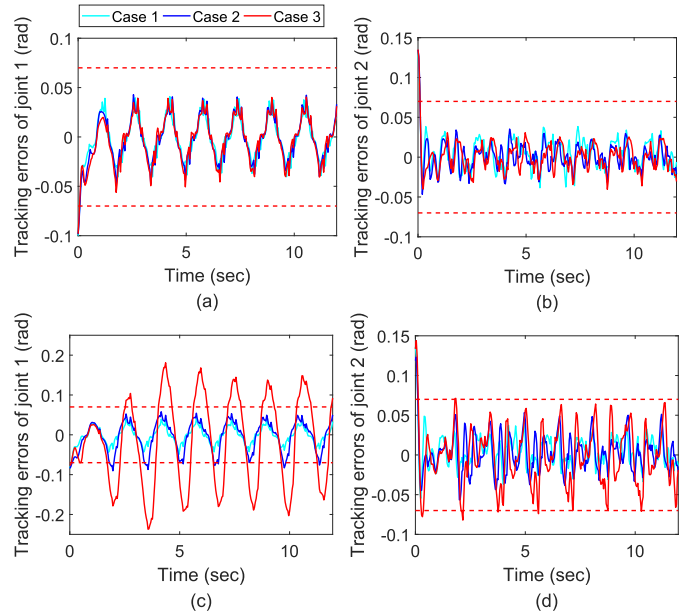


Fig. 12. Tracking performance of the proposed controller. (a) and (b) Without input saturation. (c) and (d) With input saturation.

obvious that the proposed controller outperforms the PD+ controller regardless of the input saturation and different cases with varying payloads. The proposed controller has smaller tracking errors and satisfies the output error constraints if the robot manipulator is without the input saturation effect [see Figs. 11(a) and (b) and 12(a) and (b)]. To investigate the tracking performance considering the input saturation effect, we manually set the maximum/minimum control torque as $\tau^{\max}/\tau^{\min} = \pm[5, 2]^T$ N · m. It is the value being chosen deliberately such that the input saturation would not be negligible and the constraint violation would happen under Case 2 and Case 3 [see Figs. 11(c) and (d) and 12(c) and (d)]. As shown in Fig. 12(c) and (d), both joints are in violation of

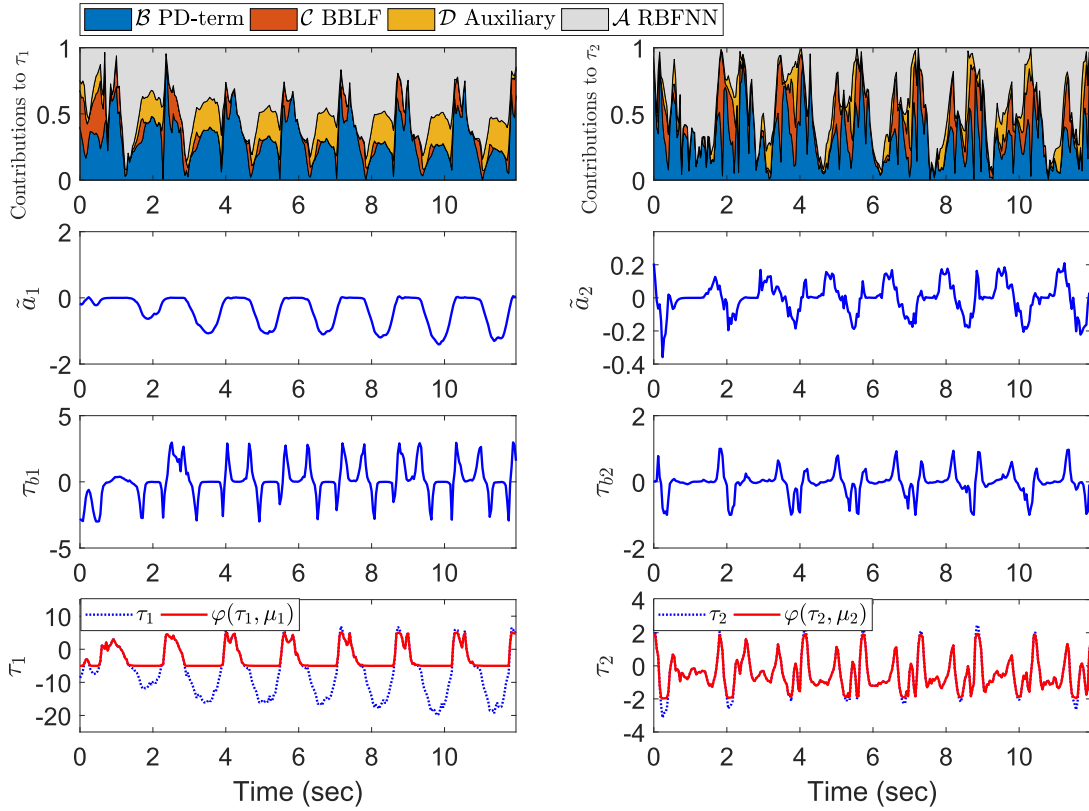


Fig. 13. Detailed tracking results of the proposed controller under Case 3.

tracking error constraints when the robot manipulator grasps a full bottle of water and tracks the predefined joint trajectories.

The detailed tracking results of the proposed controller under Case 3 are given in Fig. 13, indicating the variation of the control signals constituting the proposed controller. Compared with the proportion of $\mathcal{C}$ of τ_2 , $\mathcal{P}^c$ of τ_1 is smaller during the control. It is in line with the plots of τ_b in Fig. 2; as the constrained variable is away from the constraints, the magnitude of τ_b will become smaller. Plots of $\tilde{a}$ and τ_b show their boundedness during the control. More importantly, when the predefined constraints are not physically realizable, the proposed controller can remain stable and has smaller tracking errors compared with the PD+ controller.

VII. CONCLUSION

A trajectory tracking controller for robot manipulators is developed considering both the output error constraints and the control input saturation effects. Using the BBLF to handle the constraints' effect, we guarantee that the system could remain stable when the output errors go beyond the constrained boundaries. A modified auxiliary system and a simplified RBFNN structure are also incorporated to improve control performance. Simulation and experimental results demonstrate the validation and the superiority of the proposed method.

APPENDIX/PROOF OF THEOREM 1

Proof: Define

$$\tilde{s}(\tau, \mu) = \varphi(\tau, \mu) - \ell(\tau). \quad (26)$$

$\tilde{s}(\tau, \mu)$ is bounded, and as $\mu \rightarrow 1$, $\max(\tilde{s}(\tau, \mu)) \rightarrow 0$. Substituting (15), (18), (19), (20), and (26) into (17) yields

$$\begin{aligned} \dot{V} &= (\tau_b^T + z_1^T K_1)(z_2 + \tilde{a} - \Lambda z_1) - \text{tr}(\tilde{W}^T \Gamma^{-1} \dot{\tilde{W}}) \\ &\quad + z_2^T (W^T S + \varepsilon + H \dot{q}_r + K_a \tilde{a} - \tilde{s}(\tau, \mu) - \tau) \\ &= -(\tau_b^T + z_1^T K_1) \Lambda z_1 - z_2^T K_2 z_2 + \gamma \text{tr}(\tilde{W}^T W) \\ &\quad + (\tau_b^T + z_1^T K_1) \tilde{a} + z_2^T (\varepsilon - \tilde{s}(\tau, \mu)). \end{aligned} \quad (27)$$

Note that, using Young's inequality, there is

$$\begin{aligned} \text{tr}(\tilde{W}^T \dot{\tilde{W}}) &= \sum_{i=1}^m \sum_{j=1}^n \tilde{w}_{ij} (w_{ij} - \tilde{w}_{ij}) \\ &\leq \frac{1}{2} \sum_{i=1}^m \sum_{j=1}^n (w_{ij}^2 - \tilde{w}_{ij}^2) \\ &\leq -\frac{1}{2} \text{tr}(\tilde{W}^T \tilde{W}) + \frac{1}{2} \text{tr}(W^T W) \end{aligned}$$

and from (8) and (9), we can see that $\text{sign}(\zeta_i) = \text{sign}(z_{1i})$, $\zeta_i/z_{1i} > 0$, $h(\zeta_i)/(1 - k_{\zeta_i}^2 \zeta_i^2) > 0$, and $K^{\text{des}} > 0$. Thus, it is easy to find that $\text{sign}(\tau_b) = \text{sign}(z_1)$ and

$$\tau_b^T \Lambda z_1 = \frac{1}{2} \text{vec}^T \left(\frac{2\kappa_i (1 - \zeta_i^2) \lambda_i z_{1i}}{k_{ci} (1 - k_{\zeta_i}^2 \zeta_i^2)} \zeta_i \right) \text{vec} \left(\frac{k_i^{\text{des}} k_{\zeta_i} \zeta_i^2}{1 - k_{\zeta_i}^2 \zeta_i^2} \right).$$

Then, (27) can be rewritten as

$$\begin{aligned} \dot{V} &\leq -(\tau_b^T + z_1^T K_1) \Lambda z_1 - z_2^T K_2 z_2 - \frac{\gamma}{2} \text{tr}(\tilde{W}^T \tilde{W}) \\ &\quad + (\tau_b^T + z_1^T K_1) \tilde{a} + z_2^T (\varepsilon - \tilde{s}(\tau, \mu)) + \frac{\gamma}{2} \text{tr}(W^T W) \\ &\leq -\rho V + \varphi \end{aligned}$$

where

$$\varrho = \min \left(\frac{2\kappa_i(1 - \zeta_i^2)\lambda_i z_{1i}}{k_{ci}(1 - k_{ci}^2 \zeta_i^2)\zeta_i}, 2\lambda_{\min}(\Lambda), \frac{2\lambda_{\min}(K_2)}{\lambda_{\max}(H)}, \frac{\gamma}{\lambda_{\max}(\Gamma^{-1})} \right), \quad i = 1, 2, \dots, n$$

$$\varphi = z_2^T(\varepsilon - \tilde{s}(\tau, \mu)) + \frac{\gamma}{2}\text{tr}(W^T W) + (\tau_b^T + z_1^T K_1)\tilde{a}.$$

Note that $z_1, z_2, \varepsilon, \tilde{s}, W$, and $|\tau_b| \leq \tau^{\text{des}}$ are bounded. $\tilde{a}$ is also bounded as the auxiliary system is BIBO. Therefore, $\varphi \in L_\infty$

$$V \geq \frac{1}{2}(\lambda_{\min}(K_\zeta)\|\zeta\|^2 + \lambda_{\min}(K_1)\|z_1\|^2 + \lambda_{\min}(H)\|z_2\|^2 + \lambda_{\min}(\Gamma^{-1})\|\tilde{W}\|^2)$$

where $K_\zeta = \text{diag}((k_i^{\text{des}}k_{ci}^2\zeta_i^2)/(1 - k_{ci}^2\zeta_i^2))$. According to Lemma 1, the system is UUB stable, $t \rightarrow \infty$, $V \leq \varphi/\rho$, and the signals constructing V will converge around the small regions as follows:

$$\|\zeta\| \leq \sqrt{\frac{2\varphi}{\rho\lambda_{\min}(K_\zeta)}}, \quad \|z_1\| \leq \sqrt{\frac{2\varphi}{\rho\lambda_{\min}(K_1)}}$$

$$\|z_2\| \leq \sqrt{\frac{2\varphi}{\rho\lambda_{\min}(K_2)}}, \quad \|\tilde{W}\| \leq \sqrt{\frac{2\varphi}{\rho\lambda_{\min}(\Gamma^{-1})}}.$$

□

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